

Intelligenza Artificiale

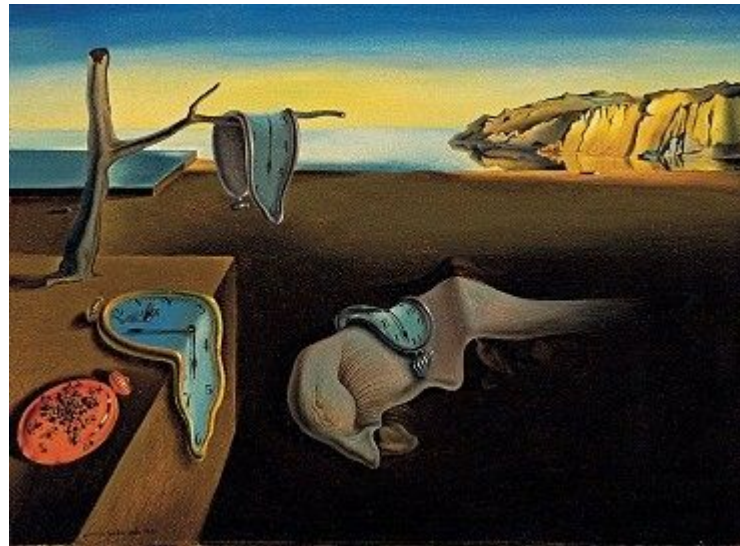
“Probabilistic reasoning over time (part 1)”

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Lecture outline

- Time and uncertainty
- Hidden Markov Models
- Inference in temporal models
 - Filtering



Slides partly based on Russell & Norvig instructors material <http://aima.cs.berkeley.edu/instructors.html>

Example: the umbrella world

- You are imprisoned in an underground jail
 - Cannot check the weather
 - Can only see whether the guard is carrying an umbrella or not
- What's the probability that outside is raining, based on today's and previous observations?
 - To answer we need reason about time...



Time and uncertainty

Assumptions

- Discrete-time models, in which the world is viewed as a series of snapshots or time slices
- The time interval Δ between slices is assumed to be the same for every interval
- $\mathbf{X}_t$: denotes the set of **state** variables at time t , assumed to be unobservable
- $\mathbf{E}_t$: denotes the set of observable **evidence** variables
- The observation at time t is $\mathbf{E}_t = \mathbf{e}_t$



Transition model

- A **transition model** specifies the probability distribution over the latest state variables, given the previous values:

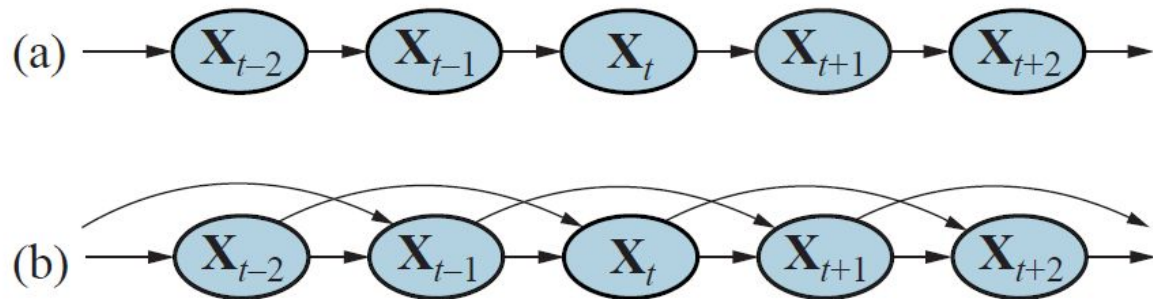
$$\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_{t-1}) \equiv \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1})$$

- but the set $\mathbf{X}_{0:t-1}$ is unbounded as t increases...
- **Markov assumption** – the current state depends only on a fixed number of previous states
 - a. **First order** Markov assumption $\rightarrow \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1}) = \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t-1})$
 - b. **Second order** Markov assumption $\rightarrow \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1}) = \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t-2}, \mathbf{X}_{t-1})$
 - ...



Andrey Andreyevich Markov
1856-1922

Markov processes

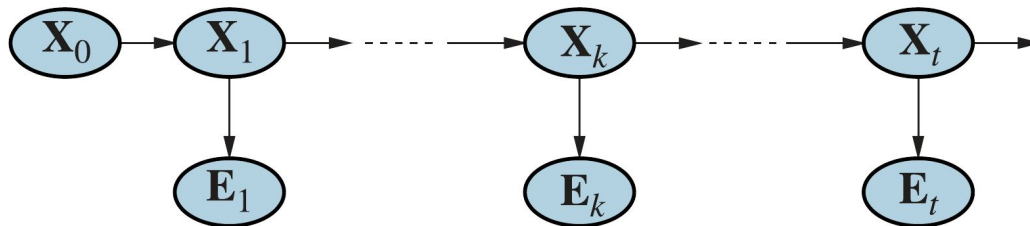


(a) Bayesian network of a **first-order Markov process**

(b) Bayesian network of a **second-order Markov process**

Hidden Markov Model (HMM)

- States $\mathbf{X}$ are not observable (hidden)
- Only observations $\mathbf{E}$ are available, and these depend on current state
- Such is the case for a (first-order) **Hidden Markov Model**, or **HMM**

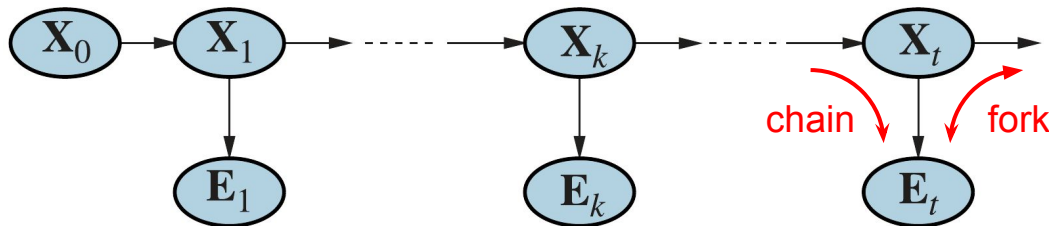


Sensor model

- The model is characterised by a **sensor Markov assumption**

$$\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{1:t-1}) = \mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$$

- That is, given $\mathbf{X}_t$, the observation $\mathbf{E}_t$ is independent from anything else
 - Recall indeed the conditional independence of chains and forks...

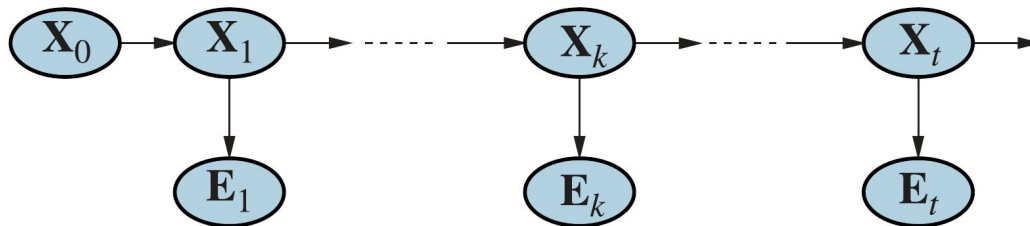


- $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$ is called the **sensor model**

Joint distribution with Markov assumptions

- Given the initial prior $\mathbf{P}(\mathbf{X}_0)$, the **full joint distribution** is

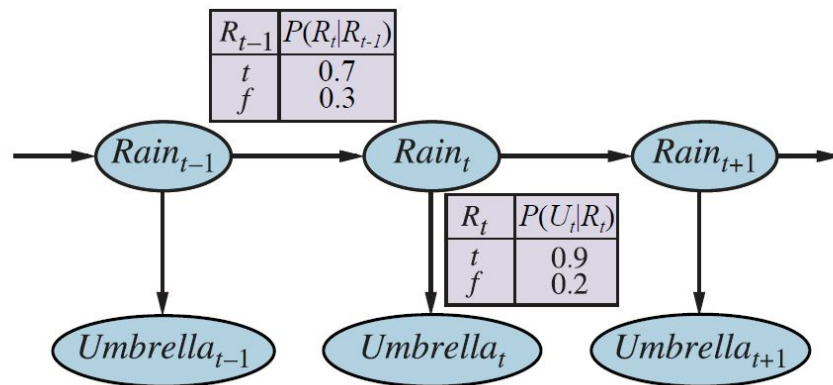
$$\mathbf{P}(\mathbf{X}_{0:t}, \mathbf{E}_{1:t}) = \mathbf{P}(\mathbf{X}_0) \prod_{i=1}^t \mathbf{P}(\mathbf{X}_i | \mathbf{X}_{i-1}) \mathbf{P}(\mathbf{E}_i | \mathbf{X}_i).$$



- $\mathbf{X}_{i-1}$ contains all the information needed to characterize the probabilities at time i

Example: umbrella world

- First-order HMMs – the probability of rain is assumed to depend only on whether it rained the previous day
- Bayesian network structure and conditional distributions:
 - **transition model** is $\mathbf{P}(\text{Rain}_t | \text{Rain}_{t-1})$
 - **sensor model** is $\mathbf{P}(\text{Umbrella}_t | \text{Rain}_t)$



Inference in temporal models



- Basic inference tasks that can be solved:
 - **Filtering** (or **state estimation**): task of computing the **belief state** $P(\mathbf{X}_t | \mathbf{e}_{1:t})$
 - **Prediction**: task of computing the posterior distribution over a future state, given all evidence to date (similar to filtering without evidence), i.e. $P(\mathbf{X}_{t+k} | \mathbf{e}_{1:t})$
 - **Smoothing**: task of computing the posterior distribution over a past state, given all evidence up to the present, i.e. $P(\mathbf{X}_{t-k} | \mathbf{e}_{1:t})$
 - **Most likely explanation**: given a sequence of observations, find the sequence of states that is most likely to have generated those observations, i.e. $\operatorname{argmax}_{\mathbf{X}_{1:t}} P(\mathbf{X}_{1:t} | \mathbf{e}_{1:t})$
- Besides inference tasks:
 - **Learning**: The transition and sensor models, if not yet known, could be learned from observations

Filtering

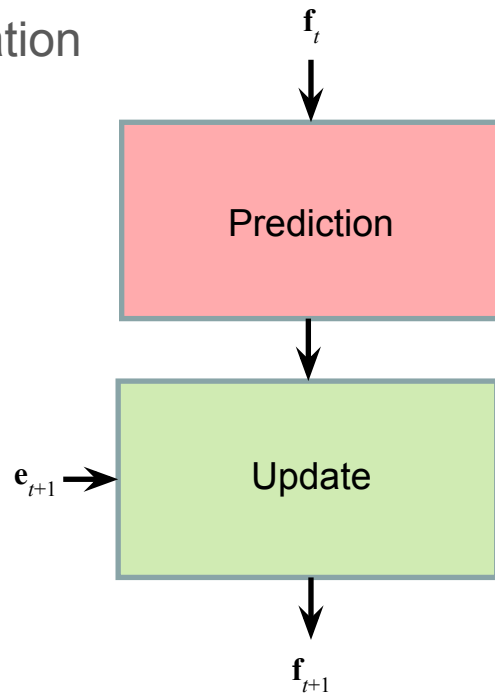
- It is a **recursive** two-steps (**prediction→update**) operation

$$\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) = \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}, \mathbf{e}_{t+1}) \quad (\text{dividing up the evidence})$$

$$= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}, \mathbf{e}_{1:t}) \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}) \quad (\text{using Bayes' rule, given } \mathbf{e}_{1:t})$$

$$\underbrace{\alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{update}} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t})}_{\text{prediction}} \quad (\text{by the sensor Markov assumption}).$$

$$\alpha = 1/P(\mathbf{e}_{t+1})$$



Filtering

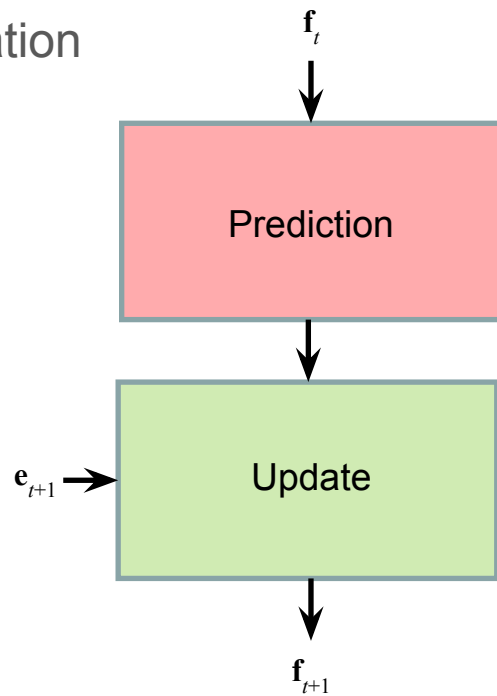
- It is a **recursive** two-steps (**prediction**→**update**) operation

$$\begin{aligned}
 \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) &= \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}, \mathbf{e}_{t+1}) \quad (\text{dividing up the evidence}) \\
 &= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}, \mathbf{e}_{1:t}) \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}) \quad (\text{using Bayes' rule, given } \mathbf{e}_{1:t}) \\
 &= \alpha \underbrace{\mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{update}} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t})}_{\text{prediction}} \quad (\text{by the sensor Markov assumption}).
 \end{aligned}$$

$$\alpha = 1/P(\mathbf{e}_{t+1})$$

- We can rewrite the prediction factor by marginalising $\mathbf{x}_t$

$$\begin{aligned}
 \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) &= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}) \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t, \cancel{\mathbf{e}_{1:t}}) P(\mathbf{x}_t | \mathbf{e}_{1:t}) \\
 &= \alpha \underbrace{\mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{sensor model}} \sum_{\mathbf{x}_t} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t)}_{\text{transition model}} \underbrace{P(\mathbf{x}_t | \mathbf{e}_{1:t})}_{\text{recursion}} \quad (\text{Markov assumption}).
 \end{aligned}$$



Filtering

- It is a **recursive** two-steps (**prediction**→**update**) operation

$$\begin{aligned}
 \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) &= \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}, \mathbf{e}_{t+1}) \quad (\text{dividing up the evidence}) \\
 &= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}, \mathbf{e}_{1:t}) \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t}) \quad (\text{using Bayes' rule, given } \mathbf{e}_{1:t}) \\
 &= \alpha \underbrace{\mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{update}} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t})}_{\text{prediction}} \quad (\text{by the sensor Markov assumption}).
 \end{aligned}$$

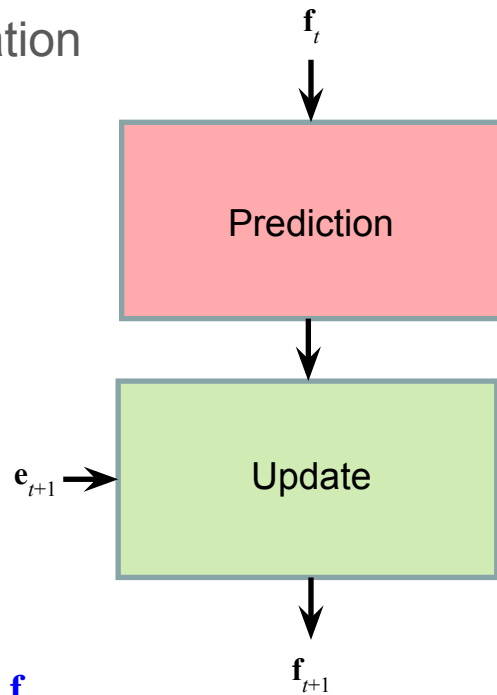
$$\alpha = 1/P(\mathbf{e}_{t+1})$$

- We can rewrite the prediction factor by marginalising $\mathbf{x}_t$

$$\begin{aligned}
 \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) &= \alpha \mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1}) \underbrace{\sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t, \cancel{\mathbf{e}_{1:t}}) P(\mathbf{x}_t | \mathbf{e}_{1:t})}_{\text{marginalising } \mathbf{x}_t} \\
 &= \alpha \underbrace{\mathbf{P}(\mathbf{e}_{t+1} | \mathbf{X}_{t+1})}_{\text{sensor model}} \sum_{\mathbf{x}_t} \underbrace{\mathbf{P}(\mathbf{X}_{t+1} | \mathbf{x}_t)}_{\text{transition model}} \underbrace{P(\mathbf{x}_t | \mathbf{e}_{1:t})}_{\text{recursion}} \quad (\text{Markov assumption}).
 \end{aligned}$$

- Basically a **FORWARD** function applied recursively to a “message” $\mathbf{f}$

$$\mathbf{f}_{t+1} \equiv \mathbf{P}(\mathbf{X}_{t+1} | \mathbf{e}_{1:t+1}) = \mathbf{FORWARD}(\mathbf{P}(\mathbf{X}_t | \mathbf{e}_{1:t}), \mathbf{e}_{t+1}) = \mathbf{FORWARD}(\mathbf{f}_t, \mathbf{e}_{t+1})$$



Filtering: umbrella world

- Two types of weather: *Rain = true* and *Rain = false* (r and $\neg r$)
- **QUERY**: what is the probability that today ($t = 2$) is raining given that you've seen an umbrella yesterday and today?

$$P(r_2 \mid u_1, u_2) = ?$$

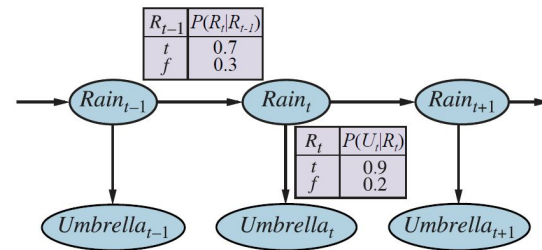
- Assuming equiprobability of the initial states at $t = 0$:

$$P(r_0) = P(\neg r_0) = 0.5$$

- Two steps filtering problem:
 1. $\mathbf{f}_1 = \text{FORWARD}(\mathbf{P}(R_0), u_1)$
 2. $\mathbf{f}_2 = \text{FORWARD}(\mathbf{f}_1, u_2)$



Filtering: umbrella world



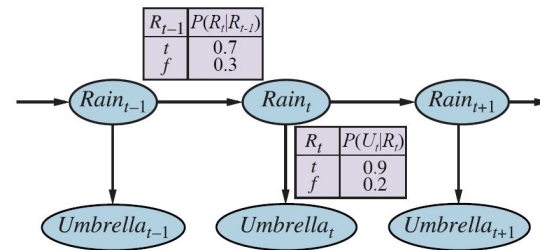
- $\mathbf{f}_1 = \text{FORWARD}(\mathbf{P}(R_0), u_1) \equiv \mathbf{P}(R_1 | u_1)$

$$\begin{aligned}
 P(r_1 | u_1) &= \alpha_1 P(u_1 | r_1) [P(r_1 | r_0) P(r_0) + P(r_1 | \neg r_0) P(\neg r_0)] \\
 &= \alpha_1 * 0.9 * [0.7 * 0.5 + 0.3 * 0.5] \\
 &= \alpha_1 * 0.9 * \mathbf{0.5} = \alpha_1 * 0.45
 \end{aligned}$$

$$\begin{aligned}
 P(\neg r_1 | u_1) &= \alpha_1 P(u_1 | \neg r_1) [P(\neg r_1 | r_0) P(r_0) + P(\neg r_1 | \neg r_0) P(\neg r_0)] \\
 &= \alpha_1 * 0.2 * [0.3 * 0.5 + 0.7 * 0.5] \\
 &= \alpha_1 * 0.2 * \mathbf{0.5} = \alpha_1 * 0.1
 \end{aligned}$$

$$\Rightarrow \mathbf{f}_1 = \alpha_1 \langle 0.45, 0.1 \rangle = \langle 0.45/0.55, 0.1/0.55 \rangle = \langle \mathbf{0.818}, \mathbf{0.182} \rangle$$

Filtering: umbrella world



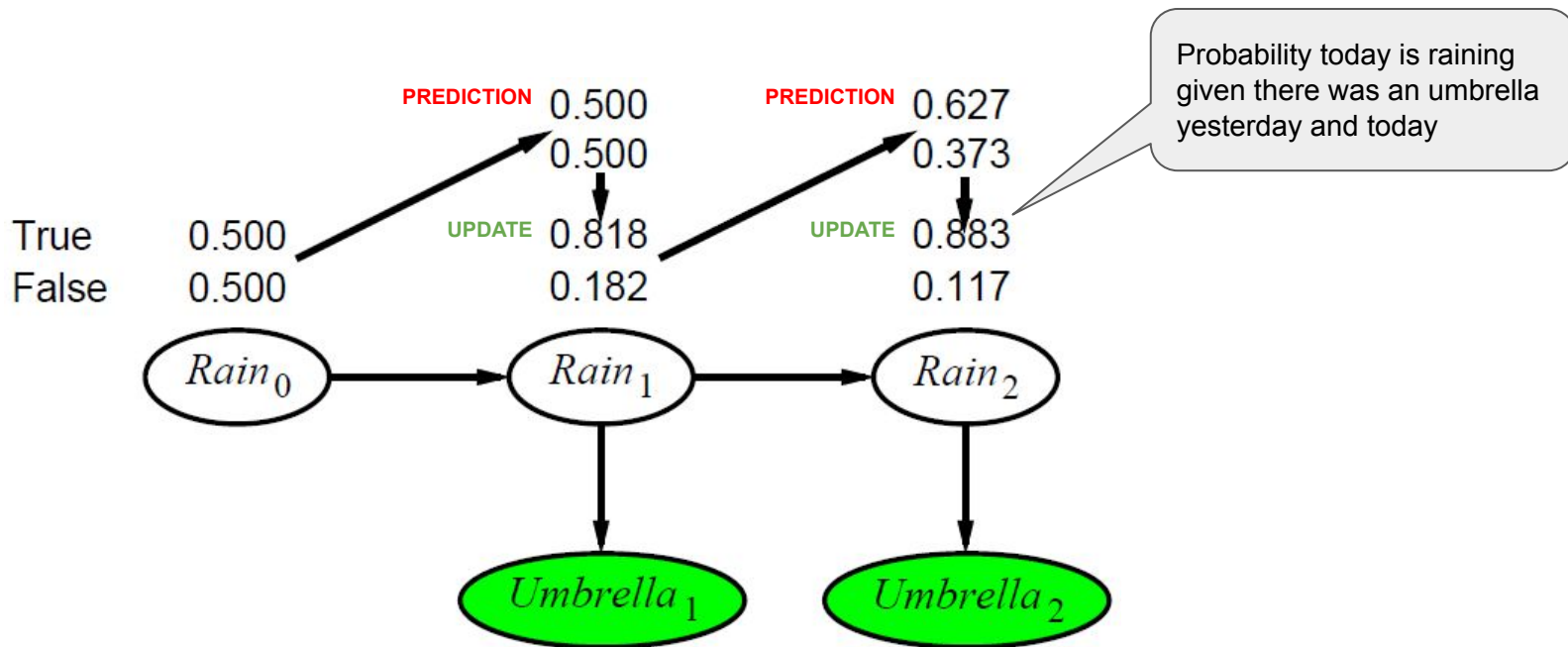
- $\mathbf{f}_2 = \text{FORWARD}(\mathbf{f}_1, u_2) \equiv \mathbf{P}(R_2 \mid u_1, u_2)$

$$\begin{aligned}
 P(r_2 \mid u_1, u_2) &= \alpha_2 P(u_2 \mid r_2) [P(r_2 \mid r_1) * \mathbf{0.818} + P(r_2 \mid \neg r_1) * \mathbf{0.182}] \\
 &= \alpha_2 * 0.9 * [0.7 * 0.818 + 0.3 * 0.182] \\
 &= \alpha_2 * 0.9 * \mathbf{0.627} = \alpha_2 * 0.564
 \end{aligned}$$

$$\begin{aligned}
 P(\neg r_2 \mid u_1, u_2) &= \alpha_2 P(u_2 \mid \neg r_2) [P(\neg r_2 \mid r_2) * \mathbf{0.818} + P(\neg r_2 \mid \neg r_2) * \mathbf{0.182}] \\
 &= \alpha_2 * 0.2 * [0.3 * 0.818 + 0.7 * 0.182] \\
 &= \alpha_2 * 0.2 * \mathbf{0.373} = \alpha_2 * 0.075
 \end{aligned}$$

$$\Rightarrow \mathbf{f}_2 = \alpha_2 \langle 0.564, 0.075 \rangle = \langle 0.564/0.639, 0.075/0.639 \rangle = \langle \mathbf{0.883}, \mathbf{0.117} \rangle$$

Filtering: umbrella world



Summary



- The changing state of the world is handled by using a set of random variables to represent the state at each point in time.
- Representations can be designed to (roughly) satisfy the **Markov property**, so that the future is independent of the past given the present.
- A temporal probability model can be thought of as containing a **transition model** describing the state evolution and a **sensor model** describing the observation process.
- Most common temporal models are (first-order) **HMMs**.
- The principal inference tasks in temporal models are **filtering** (state estimation), **prediction**, **smoothing**, and computing the **most likely explanation**.

Conclusions

- Suggested reading
 - Chapter 14, sec. 14.1-14.2 of Russell & Norvig
- Next lecture
Probabilistic reasoning over time
- Questions?

